

Build Task — Tool Library — Hints & Solution

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint has 2 depth levels: **Basic** (the plain idea) and **Intermediate** (proper LangChain/Python). Read Basic first even if you already know this — it's the fastest way to spot exactly what Intermediate adds.

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Hint 1 — What you're building, and the exact pieces

Basic Version

You're building three small files that work together: `tools.py` holds the actual tools, `tool_harness.py` runs a prompt against all of them and reports what happened, and `test_prompts.py` holds the list of prompts you'll test with.

You need exactly 2-3 tools, and they can't all be the same kind: at least one that's pure logic (no outside calls at all), one that calls a real service (reuse Doc02's HTTP client), and one built to sometimes fail on purpose (so you have something real to test the failure-handling exercise against).

Things to use:

- `@tool` from `langchain_core.tools`, with typed function parameters (which `@tool` reads automatically to build the argument shape shown to the model).
- Your Doc02 `request_with_retry()` — reuse it inside the real-API tool, don't rewrite it.
- `try/except` inside your "sometimes fails" tool, returning `f"Error: {e}"` instead of crashing.
- `model.bind_tools(ALL_TOOLS)` to register everything at once.

Try writing down, in plain words, what each of your 2-3 tools actually does, before writing any code.

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Intermediate Version

`tools.py` should export a plain list of your `@tool`-decorated functions (`ALL_TOOLS = [tool_a, tool_b, tool_c]`), so `tool_harness.py` and any later document can import that one list instead of naming each tool individually.

`tool_harness.py`'s `run_with_tools(prompt: str, tools: list) -> ToolCallResult` should be a typed function — define a small `ToolCallResult` (a dataclass is fine) that records the prompt, which tool (if any) got called, what arguments it got, and what it returned.

The "sometimes fails" tool should fail in a controllable, predictable way for testing — not randomly. A simple pattern: an argument like `should_fail: bool`, so your test harness can force the failure path on demand instead of hoping for bad luck. `test_prompts.py` should hold a list of prompts covering all three cases from the Test Cases table: a prompt that clearly matches one tool, one that matches none, and one that should trigger the failing tool.

Here's what to actually go look at for `run_with_tools()`: call `model.bind_tools(tools)`, invoke with the prompt, read `response.tool_calls`, and if one exists, find the matching tool object by name and actually run it — not just log which tool got picked.

Sketch `ToolCallResult`'s fields, and `run_with_tools()`'s full signature, before moving to Hint 2.

Difference between Basic and Intermediate: Basic names the three files and the tools' jobs in plain words. Intermediate gives each file a real, typed shape — `ToolCallResult`, a `should_fail` flag, a `run_with_tools()` that actually finds and runs the matching tool. Worth knowing this version trusts `@tool`'s auto-inferred argument shape and only reads the first tool call in a response — a real, ongoing library would eventually want an explicit Pydantic `args_schema` per tool and to handle every call in a response, not just the first, but neither is required for this Build Task as written.

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Hint 2 — The plan, and almost the whole thing

Basic Version

```
tools.py:
    a pure-logic tool (e.g. add(a, b))
    a real-API tool (calls request_with_retry)
    a sometimes-fails tool (takes should_fail, raises on purpose if true,
    caught and returned as an error string)
    ALL_TOOLS = list of the three

tool_harness.py:
    ToolCallResult holds: prompt, tool_name (or none), arguments, result

    function run_with_tools(prompt, tools):
        register tools on a model
        ask the prompt
        if no tool called: return a result with tool_name=None
        else: run the tool, return the result with its real output

test_prompts.py:
    a list of test prompts

main script:
    for each test prompt:
        result = run_with_tools(prompt, ALL_TOOLS)
        print the result
```

The trickiest part — finding the right tool object from its name, so you can actually invoke it:

```
def find_tool(tools, name):
    for candidate in tools:
        if candidate.name == name:
            return candidate
    return None
```

`@tool` -decorated functions carry a `.name` attribute (usually the function's own name). Use `find_tool()` inside `run_with_tools()`, then check against the [Solution](#).

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Intermediate Version

tools.py :

```
@tool
def add(a: int, b: int) -> int: ...           # pure logic

@tool
def get_weather(city: str) -> str: ...        # real API, via request_with_retry

@tool
def flaky_lookup(query: str, should_fail: bool = False) -> str:
    try:
        if should_fail:
            raise RuntimeError("simulated failure")
        return f"Result for {query}"
    except RuntimeError as e:
        return f"Error: {e}"

ALL_TOOLS = [add, get_weather, flaky_lookup]
```

tool_harness.py :

```
@dataclass
class ToolCallResult:
    prompt: str
    tool_name: str | None
    arguments: dict
    output: str

def find_tool(tools: list[BaseTool], name: str) -> BaseTool | None:
    for candidate in tools:
        if candidate.name == name:
            return candidate
    return None

function run_with_tools(prompt, tools) -> ToolCallResult:
    model_with_tools = ChatOpenAI(...).bind_tools(tools)
    response = model_with_tools.invoke(prompt)
    if not response.tool_calls:
        return ToolCallResult(prompt, None, {}, response.content)
```

```
call = response.tool_calls[0]
matching_tool = find_tool(tools, call["name"])
output = matching_tool.invoke(call["args"])
return ToolCallResult(prompt, call["name"], call["args"], output)
```

test_prompts.py :

```
TEST_PROMPTS = [
    "what's 5 + 7?",
    "what's the weather in Lahore?",
    "look up something but force it to fail",
    "tell me a joke",
]
```

Wire these three files together with a `main()` that loops over `TEST_PROMPTS` , then compare against the [Solution](#).

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Solution

Read both depths — they're not "wrong, right," they're 2 real, valid ways to build the same library, with real tradeoffs between them.

Basic Version

Approach 1 — one flat harness function

Story — `tools.py` : this is the one library every later document (Doc07 onward) imports instead of writing tools from scratch — 3 different kinds on purpose (pure logic, real API, deliberately fallible), so every later document has a real example of each shape to build on. **If not:** every later document would each invent its own tools, none tested against this document's failure-handling and selection-ambiguity lessons.

```
# tools.py
from langchain_core.tools import tool
from http_client import request_with_retry

@tool
def add(a: int, b: int) -> int:
    """Add two integers together and return the sum."""
    return a + b

@tool
def get_weather(city: str) -> str:
    """Get the current weather for a named city."""
    # how: reuses Doc02's request_with_retry() – the tool never calls
    # requests directly, so it gets timeout/retry handling for free.
    url = "https://api.open-meteo.com/v1/forecast"
    response = request_with_retry(url, params={"q": city})
    return f"Weather data for {city}: {response}"

@tool
def flaky_lookup(query: str, should_fail: bool = False) -> str:
    """Look up information for a query. Can be forced to fail for testing."""
    # why: should_fail makes the failure controllable and predictable for
    # testing – not something you have to hope happens on its own.
    try:
        if should_fail:
```

```

        raise RuntimeError("simulated failure")
    return f"Result for {query}"
except RuntimeError as e:
    return f"Error: {e}"

```

```
ALL_TOOLS = [add, get_weather, flaky_lookup]
```

Story — `tool_harness.py` : this is the one round trip (register tools, ask, find the matching tool, run it) every later document reuses instead of writing its own. **If not:** Doc07's agent loop would have to build this same round trip from scratch instead of wrapping this one in a `while` loop.

```

# tool_harness.py
from dataclasses import dataclass
from langchain_openai import ChatOpenAI
from tools import ALL_TOOLS

@dataclass
class ToolCallResult:
    prompt: str
    tool_name: str | None
    arguments: dict
    output: str

def run_with_tools(prompt: str, tools: list) -> ToolCallResult:
    model_with_tools = ChatOpenAI().bind_tools(tools)
    response = model_with_tools.invoke(prompt)

    if not response.tool_calls:
        # when: the model answered directly – a real, checkable outcome,
        # not something you'd otherwise have to infer from an empty dict.
        return ToolCallResult(prompt, None, {}, response.content)

    call = response.tool_calls[0]
    for candidate in tools:
        if candidate.name == call["name"]:
            output = candidate.invoke(call["args"])
            return ToolCallResult(prompt, call["name"], call["args"], output)

    error = "Error: tool not found"
    return ToolCallResult(prompt, call["name"], call["args"], error)

```

Story — `test_prompts.py` : proves which tool actually gets picked for a real set of prompts, not just an assumption — this is this Build Task's test file. **If not:** you'd ship `tools.py / tool_harness.py` never having watched them actually run against the 3 required prompt types.

```
# test_prompts.py
from tool_harness import run_with_tools
from tools import ALL_TOOLS

TEST_PROMPTS = [
    "what's 5 + 7?",
    "what's the weather in Lahore?",
    "look up something but force it to fail",
    "tell me a joke",
]

for prompt in TEST_PROMPTS:
    result = run_with_tools(prompt, ALL_TOOLS)
    print(result)
```

Expected output (example):

```
ToolCallResult(prompt="what's 5 + 7?", tool_name='add',
    arguments={'a': 5, 'b': 7}, output=12)
ToolCallResult(prompt="what's the weather in Lahore?",
    tool_name='get_weather', arguments={'city': 'Lahore'},
    output='Weather data for Lahore: ...')
ToolCallResult(prompt='look up something but force it to fail',
    tool_name='flaky_lookup',
    arguments={'query': 'something', 'should_fail': True},
    output='Error: simulated failure')
ToolCallResult(prompt='tell me a joke', tool_name=None, arguments={},
    output='Why did the chicken cross the road? ...')
```

This version works correctly and meets the Build Task's core requirements. It relies on `@tool`'s auto-inferred argument shape rather than an explicit Pydantic model, and only ever looks at the first tool call in a response — both fine for a first working version.

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Intermediate Version

Approach 1 — `find_tool()` split out, model built fresh each call

`tools.py` — identical to the Basic Version above (same 3 tools, same `ALL_TOOLS` list). Shown again here in full since this is the version the rest of this Intermediate section actually builds on:

```
# tools.py
from langchain_core.tools import tool
from http_client import request_with_retry

@tool
def add(a: int, b: int) -> int:
    """Add two integers together and return the sum."""
    return a + b

@tool
def get_weather(city: str) -> str:
    """Get the current weather for a named city."""
    url = "https://api.open-meteo.com/v1/forecast"
    response = request_with_retry(url, params={"q": city})
    return f"Weather data for {city}: {response}"

@tool
def flaky_lookup(query: str, should_fail: bool = False) -> str:
    """Look up information for a query. Can be forced to fail for testing."""
    try:
        if should_fail:
            raise RuntimeError("simulated failure")
        return f"Result for {query}"
    except RuntimeError as e:
        return f"Error: {e}"

ALL_TOOLS = [add, get_weather, flaky_lookup]
```

`tool_harness.py`

```
from dataclasses import dataclass

from langchain_core.tools import BaseTool
from langchain_openai import ChatOpenAI
```

```

@dataclass
class ToolCallResult:
    prompt: str
    tool_name: str | None
    arguments: dict
    output: str

def find_tool(tools: list[BaseTool], name: str) -> BaseTool | None:
    # why: pulled out on its own so it's testable with plain tool objects
    # and a name string – no model call needed to prove it works.
    for candidate in tools:
        if candidate.name == name:
            return candidate
    return None

def run_with_tools(prompt: str, tools: list[BaseTool]) -> ToolCallResult:
    model = ChatOpenAI(model="gpt-4o-mini", temperature=0)
    model_with_tools = model.bind_tools(tools)
    response = model_with_tools.invoke(prompt)

    if not response.tool_calls:
        return ToolCallResult(
            prompt=prompt, tool_name=None, arguments={}, output=response.content
        )

    call = response.tool_calls[0]
    matching_tool = find_tool(tools, call["name"])

    if matching_tool is None:
        error = "Error: tool not found"
        return ToolCallResult(prompt, call["name"], call["args"], error)

    output = matching_tool.invoke(call["args"])
    return ToolCallResult(
        prompt=prompt,
        tool_name=call["name"],
        arguments=call["args"],
        output=output,
    )

```

test_prompts.py

```

from tool_harness import run_with_tools
from tools import ALL_TOOLS

TEST_PROMPTS = [
    "what's 5 + 7?",
    "what's the weather in Lahore?",
    "look up something but force it to fail",
    "tell me a joke",
]

def main() -> None:
    for prompt in TEST_PROMPTS:
        result = run_with_tools(prompt, ALL_TOOLS)
        print(result)

if __name__ == "__main__":
    main()

```

Why this approach: `find_tool()` as its own function is independently testable without any model call, and `ToolCallResult`'s `tool_name=None` branch makes "no tool was needed" a real, checkable outcome rather than something you infer from an empty dict.

Approach 2 — the model is built once and reused, plus a summary count

This version separates "build the model with tools" from "run one prompt through it," so the same bound model gets reused across every test prompt instead of rebuilding it each time — cheaper, and closer to how you'd actually run a batch of test prompts in real code.

```

# tool_harness.py
def build_model_with_tools(tools: list[BaseTool]) -> ChatOpenAI:
    model = ChatOpenAI(model="gpt-4o-mini", temperature=0)
    return model.bind_tools(tools)

def run_with_tools(
    prompt: str,
    tools: list[BaseTool],
    model_with_tools: ChatOpenAI | None = None,
) -> ToolCallResult:
    # why: model_with_tools defaults to None so a batch of test prompts can
    # build it once and reuse it, instead of rebuilding on every call.
    if model_with_tools is None:

```

```

    model_with_tools = build_model_with_tools(tools)

    response = model_with_tools.invoke(prompt)

    if not response.tool_calls:
        return ToolCallResult(prompt, None, {}, response.content)

    call = response.tool_calls[0]
    matching_tool = find_tool(tools, call["name"])
    if matching_tool is None:
        error = "Error: tool not found"
        return ToolCallResult(prompt, call["name"], call["args"], error)

    output = matching_tool.invoke(call["args"])
    return ToolCallResult(prompt, call["name"], call["args"], output)

```

```

# test_prompts.py
from tool_harness import build_model_with_tools, run_with_tools
from tools import ALL_TOOLS

TEST_PROMPTS = [
    "what's 5 + 7?",
    "what's the weather in Lahore?",
    "look up something but force it to fail",
    "tell me a joke",
]

def main() -> None:
    model_with_tools = build_model_with_tools(ALL_TOOLS)
    triggered = 0

    for prompt in TEST_PROMPTS:
        result = run_with_tools(prompt, ALL_TOOLS, model_with_tools)
        print(result)
        if result.tool_name is not None:
            triggered += 1

    print(f"\n{triggered}/{len(TEST_PROMPTS)} prompts triggered a tool")

if __name__ == "__main__":
    main()

```

Difference from Basic: both Intermediate approaches add `find_tool()` as an independently testable function, and full type hints throughout. Approach 2 additionally separates building the bound model from running a prompt through it, so a batch of test prompts reuses one model instead of rebuilding it per prompt, and adds a summary line (`N/M triggered a tool`) — useful once this becomes a real regression check instead of a one-off look.

Which one should you actually use? For the Build Task as written, Intermediate Approach 2 already satisfies every stated requirement and constraint, and is what most people should ship first. Two gaps are worth naming rather than fixing here: this version trusts `@tool` 's auto-inferred argument shape instead of an explicit Pydantic `args_schema` , and `run_with_tools()` only ever reads the first tool call in a response, silently dropping a second one if a prompt ever triggers two calls at once. Neither is required for this Build Task, but both are worth watching for as this library gets reused in later documents.